

# Factored Interaction Fields: Learning Joint-Anchored Hand-Object Proximity Through Its Geometric Arguments

Tanvir Ahmed<sup>a,\*</sup>

<sup>a</sup>North South University, Dhaka, Bangladesh

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## Abstract

The interaction field of the SHOW3D benchmark assigns to each of the 21 joints of a hand the three-dimensional vector to the nearest point of the manipulated object, observed from the two cameras of an egocentric headset. Published solutions regress these 126 numbers directly, although the field is a deterministic function of three supervised quantities: the hand joints, the object pose and a known object mesh. We propose factored interaction fields (FIF), a transformer decoder over frozen DINOv3 tokens that predicts camera-frame joints and the object pose, evaluates the field through a differentiable soft nearest-vertex operator on the posed mesh, and fuses this geometric estimate with a direct regression head by predicted precision. On held-out subjects, FIF reduces the error of a matched direct regressor from 46.2 to 44.4 mm and from 35.7 to 34.1 mm on a second subject pair, against 60.5 mm for the official baseline; a single-view FIF outperforms the two-view direct model. The gain grows with difficulty, from 0.6 mm for joints within 15 mm of the object to 21 mm when the hand leaves the field of view. Inside the model the improvement is carried by the direct head, whose representation is shaped by the geometric supervision; inference-time fusion contributes 0.1 mm and the geometric head’s predicted scale saturated in four of five runs. The advantage does not carry over to HOT3D in four cross-dataset settings, which we trace to the object-pose branch. We release code, derived labels and the HOT3D-IF cross-dataset protocol.

**Keywords:** hand-object interaction, interaction field, egocentric vision, 3D hand pose, 6-DoF object pose, uncertainty-weighted fusion

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## 1. Introduction

How close each part of a hand is to the object it manipulates, before, during and after contact, is a compact description of manipulation that robot learning, augmented-reality interfaces and activity understanding can consume directly. ARCTIC [1] formalised this quantity as an *interaction field*, the distance from every point of one shape to the closest point of the other, and TACO [2] extended it to tool use. SHOW3D [3] brought the problem into the wild with 4.3 M frames of hands manipulating 21 objects in more than 30 indoor and outdoor locations, annotated in three dimensions by an ego-exo capture rig. Its HANDS@ECCV 2026 challenge asks for a joint-anchored version of the field: for each of the 21 UmeTrack landmarks [4] of each hand, the three-dimensional vector to the nearest vertex of the object mesh, estimated from two synchronised egocentric views of a Meta Quest 3 headset [5] (Fig. 1).

The challenge produced strong systems. GOLF [6] adapts a DINOv3 ViT-H+ [7] with low-rank adaptation (LoRA) [8], samples region tokens around the hands and the object, tags every token with the Plücker coordinates of its viewing ray and decodes 42 joint queries into vectors; JSSR [9] adds temporal context and an epipolar search for the field endpoints. Between the official ResNet-50 baseline and GOLF, the hidden-test error

fell from 59 to 28 mm. GOLF’s own submission trace is instructive: most of the gain came from the backbone (5.7 mm from ViT-L to the adapted ViT-H+) and from local detail (2.7 mm), whereas adding the second view changed the error by 0.19 mm and ray encoding by 0.6 mm [6]. With a 64 mm baseline and hands at 200 to 400 mm, one pixel of disparity corresponds to 1.4 to 5.5 mm of depth, so stereo is geometrically informative; the small empirical gain suggests either that the remaining error is not primarily a depth error, or that direct regression does not exploit the geometry it is given. Nobody has reported where the remaining 28 mm come from.

This paper starts from an observation about the target rather than the architecture. The field is a deterministic function of three quantities: the hand joints, the six-degree-of-freedom pose of the object, and the object mesh. All three are supervised in the training set, the mesh is a fixed asset shipped with the benchmark, and the object category is provided at test time. Yet every published solution regresses the 126 numbers directly and ignores that the 21 endpoints of a hand lie on one rigid surface. The same structure has long been exploited in the contact literature, where proximity is derived from posed hand and object geometry rather than estimated from pixels [10, 11, 12, 13, 14]; the interaction field is exactly such a derived quantity.

We therefore propose *factored interaction fields* (FIF): a transformer decoder over frozen DINOv3 tokens of both views that predicts camera-frame hand joints and the object pose, evaluates the field through a differentiable soft nearest-vertex operator on the posed mesh, and fuses this geometric estimate with a

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\*Corresponding author.

Email address: tanvir.ahmed32@northsouth.edu (Tanvir Ahmed)

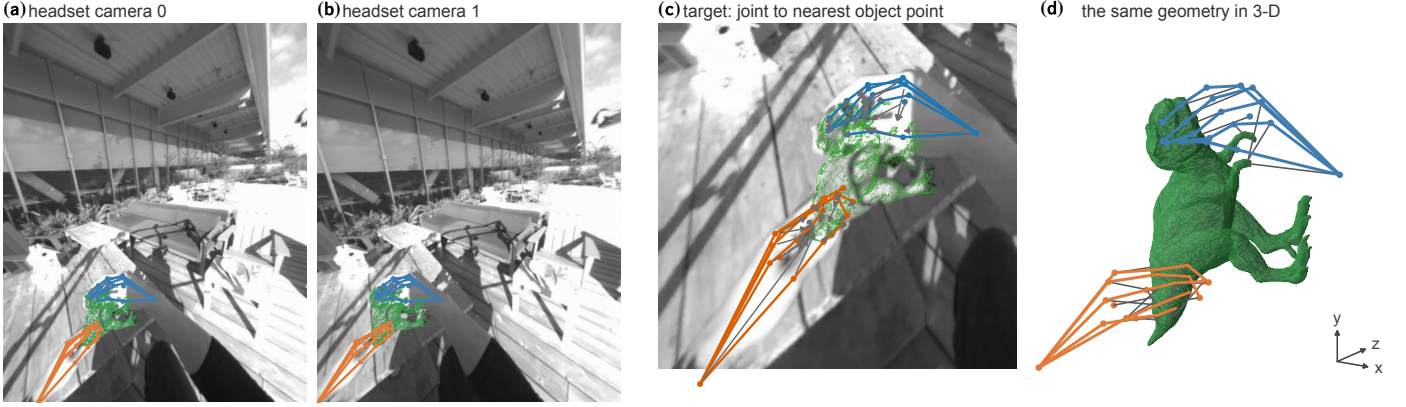


Figure 1: The joint-anchored interaction field on a held-out SHOW3D frame. (a, b) The two synchronised headset views with the ground-truth skeletons (left hand orange, right hand blue) and the posed object mesh (green wireframe). (c) The reference view cropped around the interaction; every grey arrow is the target vector from a joint to the nearest point of the posed mesh. (d) The same joints, mesh and vectors in three dimensions: the 21 endpoints of each hand lie on one rigid surface, which is the structure the proposed model exploits.

direct regression head using per-joint predicted precisions. The geometric head yields fields whose endpoints lie on the object by construction and is invariant to object symmetries; the direct head remains a fallback where the pose is not observable; the fusion is a likelihood-weighted average rather than a learned gate or an ensemble. Because the factored model exposes its intermediate quantities, it also enables an analysis the field itself does not permit: swapping predicted and ground-truth joints or object poses decomposes the error into hand and object components, and stratifying by field magnitude, joint visibility and observability shows which frames drive the mean.

We evaluate under a deliberately controlled protocol. The evaluation server is closed and the test labels are withheld, so all experiments use subject-level splits of the training subjects; all token-based models share one frozen backbone, one feature cache, one decoder and one optimiser, so that differences are attributable to the factorisation alone; the main comparison is repeated on two disjoint pairs of held-out subjects and with two seeds; and the falsification criterion was fixed before any model was trained: the hypothesis is rejected if the factored model does not beat the direct head on the mean error at a matched representation, *and* is not better on far-field or occluded joints, *and* does not transfer better to a second dataset. We also build HOT3D-IF, a joint-level version of the task on HOT3D [15], whose objects, hand skeleton and headset SHOW3D shares, by rectifying its fisheye streams and recovering landmarks through UmeTrack forward kinematics.

The results support the first two clauses and refute the third, and the analysis of the trained models turns out to be more informative than the headline number. FIF lowers the error of the matched direct regressor from 46.2 to 44.4 mm on the official development subjects (two seeds each, ranges not overlapping) and from 35.7 to 34.1 mm on a second subject pair, against 60.5 mm for the official baseline. A single-view FIF at 45.2 mm outperforms the two-view direct model. The gain rises with difficulty, from 0.6 mm for joints within 15 mm of the object to 3.0 mm beyond 60 mm and 21 mm when the hand leaves the

image. Inside the model, however, the improvement is carried by the direct head, whose representation is shaped by the geometric supervision: inference-time fusion contributes 0.1 mm, the geometric head alone is 11 mm worse than the direct head, and its predicted scale saturated at the clamp in four of the five precision-fusion runs we inspected. The object pose, not the hand, is the bottleneck of the geometric route in domain, and the hand becomes the bottleneck out of domain. The advantage does not transfer to HOT3D in any of four cross-dataset settings and disappears when HOT3D frames are added to training. We report these findings as they are, because they bound what a factored representation can be expected to deliver and where the remaining error actually sits.

Our contributions are: (i) a factored formulation of joint-anchored interaction fields with a differentiable nearest-vertex operator, closed-form differentiable object pose and precision-weighted fusion with direct regression (Section 4); (ii) a controlled evaluation at a fixed frozen representation on two held-out subject pairs, with a stratified analysis showing that the gain concentrates in far-field, occluded and out-of-view joints (Section 6); (iii) an analysis of the trained models, including oracle swaps of the geometric arguments and the calibration of the learned precisions, which locates the gain in the shared representation rather than in the fusion (Section 6.4); (iv) HOT3D-IF, a joint-level cross-dataset protocol between SHOW3D and HOT3D, and the finding that the factored advantage does not transfer (Section 6.5). All experiments run on a single laptop with 16 GB of unified memory; code, configurations, derived labels and every trained checkpoint are released.

## 2. Related work

### 2.1. Interaction fields

ARCTIC [1] introduced interaction-field estimation as the dense, per-vertex distance from one mesh to the closest vertex of another, and proposed InterField, a PointNet regressor over

canonical-pose vertices concatenated with a global image feature, in single-frame and recurrent variants. TACO [2] extended the benchmark to tool-object fields. SHOW3D [3] re-used the scalar formulation to study transfer between in-the-wild and studio data, and its challenge starter kit [5] defined the joint-anchored three-dimensional vector field of Eq. (1) together with an image-only ResNet-50 [16] baseline. The published challenge solutions share one architecture: DINOv3 tokens from both headset views decoded by 42 joint queries into vectors trained with the endpoint error. GOLF [6] adapts a ViT-H+ with LoRA [8], samples region tokens around localised hands and the object, and tags tokens with Plücker rays; JSSR [9] adds temporal context, an epipolar candidate search for the endpoints and a learned residual gate. None of them uses the object mesh or predicts the quantities that define the field; GOLF uses the object category only to condition its region locator, and JSSR deliberately avoids CAD models. Our work is complementary to backbone adaptation and region sampling and targets the missing structure.

## 2.2. Contact and proximity from posed geometry

Contact maps, potential fields and signed distances are routinely derived from posed hand and object meshes. CPF [10] models contacting vertex pairs as springs, ContactOpt [11] refines grasps towards a predicted contact map, DECO [17] predicts dense body-scene contact, HOISDF [12] and gSDF [13] constrain joint hand-object pose regression with signed distance fields, and EgoPHI [14] estimates dense contact and force on meshes from a single egocentric image given the object geometry. Datasets such as ContactPose [18] and GRAB [19] record contact as a consequence of pose. These methods treat proximity as derived from geometry; the interaction field is exactly such a derived quantity, and our geometric head makes the dependence explicit and differentiable. A recent survey of hand-object interaction with foundation models [20] distinguishes contact maps, affordance maps, field representations and dense correspondences, and flags the gap between geometric completeness and interaction correctness that motivates our decomposition.

## 2.3. Egocentric hand and hand-object pose

Egocentric benchmarks with three-dimensional hand and object annotations have grown from single-view RGB-D [21] through multi-camera studio capture [22, 23, 24, 25] to headset capture at scale [26, 15, 27, 3]. Hand-object reconstruction from a single image was addressed by ObMan [28]; large vision transformers established strong single-hand mesh recovery [29]; recent egocentric hand-object pose estimators such as HOP-former [30] and EgoGrasp [31], and the HANDS 2023 challenge analysis [32], report that close contact, narrow object views and fast motion remain the hard cases. Most of these datasets use the MANO hand model [33], whose joint convention differs from the UmeTrack landmarks of SHOW3D and HOT3D by several millimetres; we therefore restrict cross-dataset experiments to HOT3D, the only public dataset that shares the skeleton, the meshes and the headset.

## 2.4. Object pose

Our geometric head recovers the object pose in closed form from predicted control points by weighted Procrustes alignment [34, 35, 36] and is trained with the symmetry-agnostic ADD-S distance of PoseCNN [37], the mean distance from each posed model point to the closest point of the ground-truth posed model, which the BOP benchmark adopted [38]. SHOW3D’s own object ground truth was produced with FoundPose [39] followed by tracking, with reported translation errors of 1.5 to 3.5 mm [3], a noise floor relevant to our oracle analysis.

## 2.5. Multi-view geometry in networks

Epipolar Transformers [40] and learnable triangulation [41] fuse features across calibrated views for human pose; geometry-biased attention [42] is a related alternative; UmeTrack [4] is the multi-view, world-space egocentric hand tracker whose skeleton SHOW3D and HOT3D adopt. Ray embeddings introduced for neural rendering [43] and view synthesis [44] let a transformer reason about camera geometry without explicit epipolar operations; GOLF and our baselines use them. The challenge reports found stereo fusion to add little to direct regression [6, 9]; our decomposition of the endpoint error into a component along the viewing ray and a lateral remainder examines why.

## 2.6. Soft nearest neighbours and differentiable geometry

Replacing a hard argmin by a temperature-controlled softmax is a standard device for making discrete geometric operations differentiable, used for disparity regression [45] and for neural nearest-neighbour selection [46]. Our soft nearest-vertex operator applies the same idea to the nearest-surface-point map and anneals the temperature so that the exact operator is recovered at test time.

## 2.7. Uncertainty and fusion

Heteroscedastic regression, in which a network predicts the variance of its own residual and is trained with a Gaussian negative log-likelihood, dates back to Nix and Weigend [47] and was brought to computer vision by Kendall and Gal [48]; its calibration is known to be imperfect [49], and aleatoric heads have been shown to improve three-dimensional hand pose [50]. Combining several estimators by their predicted precisions is the maximum-likelihood rule for independent Gaussians and is closely related to mixtures of experts [51] and deep ensembles [52]. We use per-joint heteroscedastic scales both as a training objective and as the fusion rule between two estimators of the same field, and we report how the learned scales behaved in practice, including a run in which one head’s scale saturated.

## 2.8. Transfer between datasets

That models fit the peculiarities of the dataset they were trained on has been documented since Torralba and Efros [53]; SHOW3D itself reports scalar-field transfer between studio and in-the-wild data [3]. We add a joint-level transfer protocol between SHOW3D and HOT3D and test whether geometry-factored predictors transfer better than appearance-driven regressors.

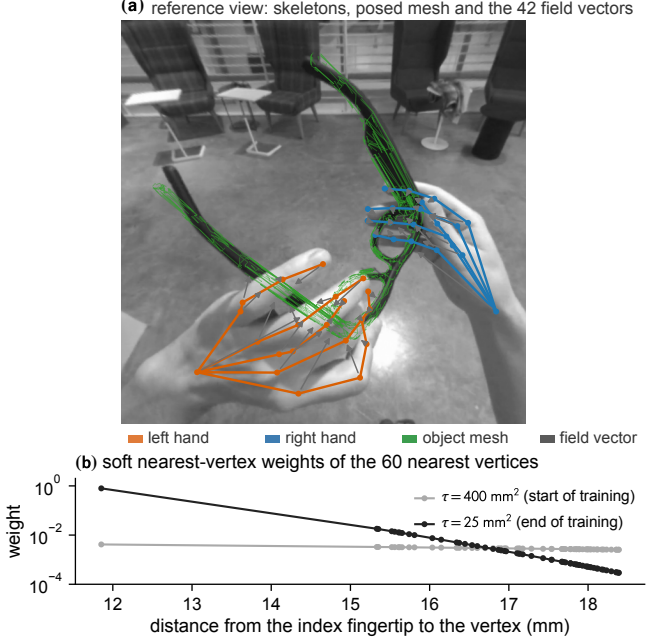


Figure 2: Geometry of the target on a SHOW3D training frame (subject ASC023 inspecting the glasses). (a) Projected skeletons, posed object mesh and the 21 nearest-vertex vectors of each hand in the reference view. (b) The soft nearest-vertex weights of Eq. (7) for the index fingertip at the initial and final training temperature: at  $\tau=400 \text{ mm}^2$  the weight is spread over the local surface patch, at  $\tau=25 \text{ mm}^2$  it is concentrated on the nearest vertex.

### 3. Problem formulation

*Task.* A SHOW3D sample is one time instant of an egocentric recording observed by two synchronised, calibrated headset cameras  $v \in \{0, 1\}$  with pinhole intrinsics  $\mathbf{K}_v$  and world-from-camera transforms  $\mathbf{T}_{w \leftarrow v} = [\mathbf{R}_{w \leftarrow v} | \mathbf{t}_{w \leftarrow v}]$  [3, 5]. For each hand  $h \in \{L, R\}$  with joints  $j = 1, \dots, J$  ( $J=21$  UmeTrack landmarks [4]), the target is the vector from the joint to the nearest vertex of the manipulated object,

$$\mathbf{v}_{h,j} = \mathbf{q}_{m^*(h,j)}(\theta_o) - \mathbf{p}_{h,j}, \quad (1)$$

$$m^*(h, j) = \arg \min_m \|\mathbf{q}_m(\theta_o) - \mathbf{p}_{h,j}\|_2. \quad (2)$$

where  $\mathbf{p}_{h,j} \in \mathbb{R}^3$  is the joint position,  $\mathbf{Q}_a = \{\mathbf{q}_m\}_{m=1}^{M_a}$  is the canonical mesh of the object with alias  $a$ , and  $\mathbf{q}_m(\theta_o) = \mathbf{R}_o \mathbf{q}_m + \mathbf{t}_o$  are its vertices under the rigid pose  $\theta_o = (\mathbf{R}_o, \mathbf{t}_o)$ . Predictions are two  $21 \times 3$  arrays per frame in millimetres, expressed in the world frame. The world frame of SHOW3D is the frame of a body-mounted rig and changes over time, but the field is a difference of two points and is therefore translation invariant, so only the per-frame rotation  $\mathbf{R}_{w \leftarrow c}$  of a reference camera  $c$  is needed to move between camera and world frames. The object alias  $a$  is part of the released test manifest and was used by the winning challenge entry [6]; we treat it as an input. Figure 2 illustrates the geometry on a real frame.

*Evaluation.* The official evaluator reports, per directed field, the mean per-joint Euclidean endpoint error (ADE) over valid targets, the fraction of joints within 10, 50 and 100 mm, and recall, the fraction of valid targets that received a prediction.

A target is valid when the object pose and the hand pose are both confident (confidence above 0.5) and headset tracking succeeded [5]. We report the same quantities on held-out subjects and add the geometric and stratified measures of Section 5. Our re-implementation of the evaluator reproduces the organisers' code exactly.

*The field is a derived quantity.* Equation (1) defines  $\mathbf{v}_h = G(\mathbf{p}_h, \theta_o; \mathbf{Q}_a)$  as a deterministic function of the hand joints, the object pose and a known mesh. Published solutions learn the composition image  $\mapsto \mathbf{v}$  end to end with a joint-query decoder [6, 9]; the geometric structure of  $G$ , in particular that all 21 endpoints of a hand lie on one rigid surface, is not used. Our method estimates the arguments of  $G$  and applies  $G$  itself. Two consequences follow. First, a prediction made through  $G$  has endpoints on the object surface by construction, whatever the error in the arguments. Second, because  $G$  depends on the posed surface only, it is invariant to the symmetry group of the object: the rotation of a can about its axis changes  $\theta_o$  but not  $G$ , so the object-pose branch need not resolve symmetric ambiguities that the field does not care about.

*Falsification criterion.* Before any model was trained we fixed three predictions that the factored formulation has to meet, all at a matched frozen representation: (i) the factored model beats the direct head on mean ADE; (ii) the geometric supervision helps most on far-field and occluded joints; and (iii) transfer to a second dataset improves. The hypothesis is refuted if all three fail. The three clauses are tested in Sections 6.1, 6.2 and 6.5 respectively.

### 4. Method

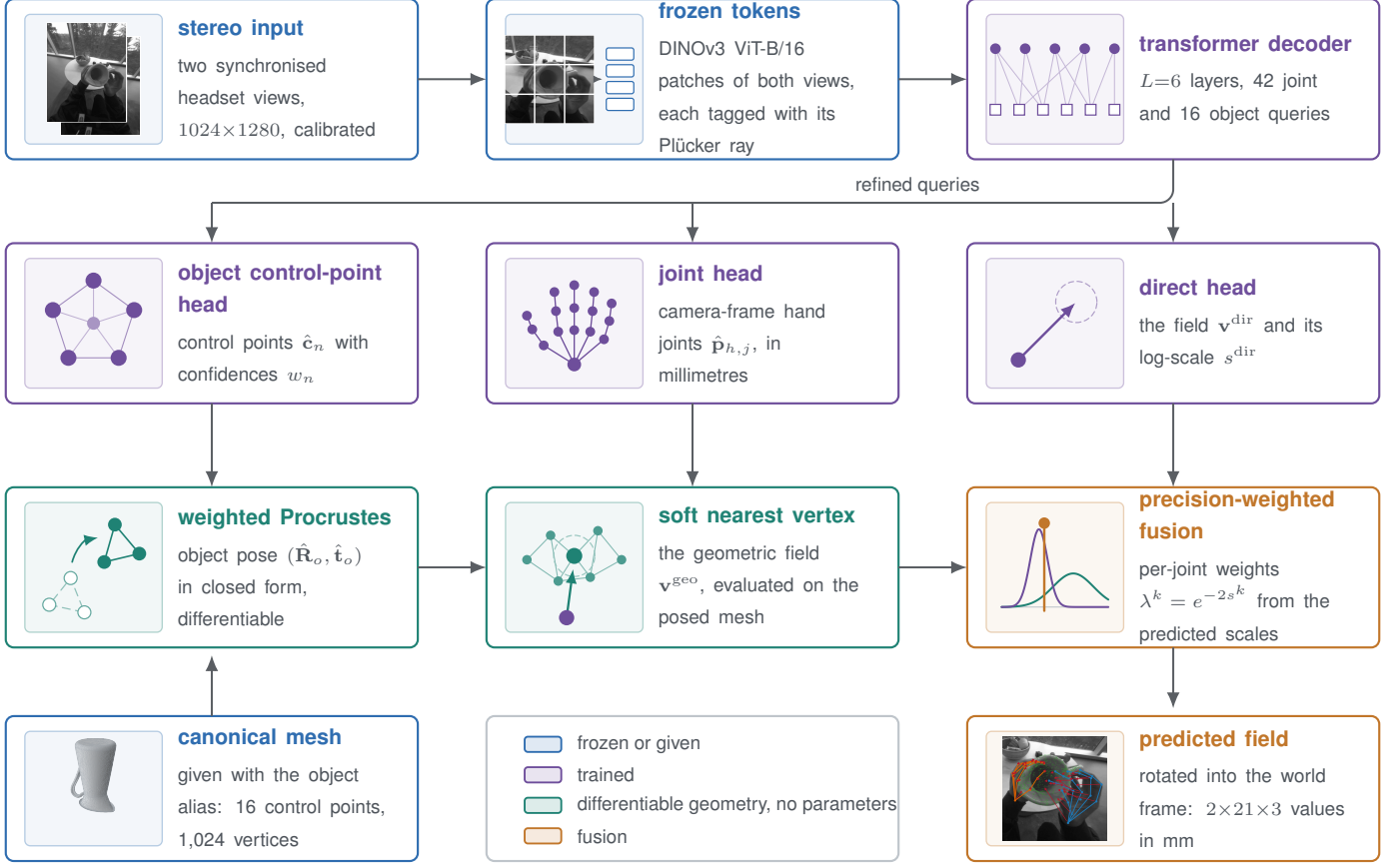
Figure 3 summarises FIF. A frozen encoder produces camera-aware tokens for both views; a transformer decoder with joint and object queries drives two complementary estimators of the field, a direct regression head and a geometric head that predicts hand joints and the object pose and evaluates Eq. (1) through a differentiable nearest-vertex operator; the two are fused by their predicted precisions.

#### 4.1. Camera-aware frozen tokens

Each view is resized to  $480 \times 384$  and encoded by a frozen DINOv3 ViT-B/16 [7, 54, 55], yielding  $N=720$  patch tokens  $\mathbf{X}_v \in \mathbb{R}^{N \times D}$  per view. Following GOLF [6], every token is tagged with the Plücker coordinates  $\pi(u, v) = [\mathbf{d}, \mathbf{o} \times \mathbf{d}] \in \mathbb{R}^6$  of its pixel ray expressed in the reference camera, with unit direction  $\mathbf{d} = \text{normalize}(\mathbf{R}_{c \leftarrow v} \mathbf{K}_v^{-1} [u; 1])$  and origin  $\mathbf{o}$  in metres [43], so that stereo geometry enters the network explicitly:

$$\mathbf{X}'_v = \mathbf{P} \mathbf{X}_v + \text{MLP}(\pi) + \mathbf{e}_{\text{view}}(v) + \mathbf{e}_{\text{pos}}. \quad (3)$$

Here  $\mathbf{P}$  projects the cached tokens to the decoder width,  $\mathbf{e}_{\text{view}}$  is a learned view embedding and  $\mathbf{e}_{\text{pos}}$  a learned positional embedding over the  $30 \times 24$  token grid. The memory is the concatenation of both views' tokens (1440 tokens; 720 for single-view variants). Because the encoder is frozen, tokens are computed once and cached, which is what makes the study feasible on modest hardware; the design is agnostic to the encoder.



**training:** the endpoint error of the fused field, a Gaussian negative log-likelihood per head, and auxiliary losses on the three heads

Figure 3: Overview of the factored interaction-field model, drawn on a held-out frame. First row: both views are encoded once by a frozen backbone and cached, every patch token is tagged with the Plücker coordinates of its pixel ray, and a decoder refines 42 joint queries and 16 object queries against the tokens of both views. Second row: three linear heads read those queries; the direct head regresses the field itself, while the joint head and the object control-point head supply the two arguments of  $G$ . Third row, each item under the head that feeds it: weighted Procrustes alignment turns the predicted control points into an object pose in closed form, the soft nearest-vertex operator evaluates the field on the mesh posed by it, so that every predicted endpoint lies on the object surface by construction, and the two field estimates are averaged with weights given by their own predicted precisions. Fourth row: the canonical mesh, which is given with the object alias, the colour key, and the output after rotation into the world frame. The auxiliary losses train the heads on frames that the field loss cannot use.

#### 4.2. Decoder and heads

Learned queries comprise  $2J=42$  joint queries  $\mathbf{z}_{h,j}$  and  $N_o=16$  object queries  $\mathbf{z}_n^o$ , one per canonical control point  $\mathbf{c}_n \in Q_a$  selected by farthest-point sampling [56]; the object alias enters through an embedding added to the object queries. An  $L$ -layer decoder [57] with self-attention over all queries and cross-attention to the memory refines them; the joint-query formulation follows DETR-style set prediction [58] as adapted for hands by the winning ARCTIC entry [59]. Linear heads then predict the direct field and its log-scale, the camera-frame joints, and the object control points with confidences,

$$\mathbf{v}_{h,j}^{\text{dir}} = \mathbf{W}_v \mathbf{z}_{h,j}, \quad s_{h,j}^{\text{dir}} = \mathbf{w}_s^\top \mathbf{z}_{h,j}, \quad \hat{\mathbf{p}}_{h,j} = \mathbf{W}_p \mathbf{z}_{h,j}, \quad (4)$$

$$\hat{\mathbf{c}}_n = \mathbf{W}_c \mathbf{z}_n^o, \quad w_n = \text{softmax}_n(\mathbf{w}_w^\top \mathbf{z}_n^o), \quad (5)$$

together with per-hand presence and per-joint visibility logits used only as auxiliary targets. Regression outputs are scaled by 100 mm and joint predictions are offset by 300 mm in depth so that the heads start near the data scale. Log-scales are clamped

to  $[-3, 5]$ , corresponding to standard deviations between 0.05 and 148 mm; the upper bound turns out to matter in Section 6.4.

#### 4.3. Geometric field

The object pose is obtained in closed form from the predicted control points by weighted Procrustes alignment [34, 35, 36],

$$(\hat{\mathbf{R}}_o, \hat{\mathbf{t}}_o) = \arg \min_{\mathbf{R} \in SO(3), \mathbf{t}} \sum_n w_n \|\mathbf{R} \mathbf{c}_n + \mathbf{t} - \hat{\mathbf{c}}_n\|^2, \quad (6)$$

which is differentiable through the singular value decomposition of a  $3 \times 3$  matrix. The posed vertices  $\hat{\mathbf{q}}_m = \hat{\mathbf{R}}_o \mathbf{q}_m + \hat{\mathbf{t}}_o$  and the predicted joints define a soft nearest-vertex field,

$$w_{h,j,m} = \frac{\exp(-\|\hat{\mathbf{q}}_m - \hat{\mathbf{p}}_{h,j}\|^2/\tau)}{\sum_{m'} \exp(-\|\hat{\mathbf{q}}_{m'} - \hat{\mathbf{p}}_{h,j}\|^2/\tau)}, \quad (7)$$

$$\mathbf{v}_{h,j}^{\text{geo}} = \sum_m w_{h,j,m} \hat{\mathbf{q}}_m - \hat{\mathbf{p}}_{h,j}, \quad (8)$$

with a temperature  $\tau$  (in  $\text{mm}^2$ ) annealed geometrically from 400 to 25 over training and replaced by the exact minimum at test

time, which reproduces Eq. (1). The soft assignment is the same device that makes disparity regression [45] and neural nearest-neighbour selection [46] differentiable. The two consequences of Section 3 now hold for the prediction itself: every endpoint  $\hat{\mathbf{p}}_{h,j} + \mathbf{v}_{h,j}^{\text{geo}}$  lies on the predicted object surface by construction, and the estimate inherits the symmetry invariance of  $G$ , which is why the auxiliary pose loss below is symmetry-agnostic. A second log-scale  $s_{h,j}^{\text{geo}}$  is predicted by a two-layer network from the joint query, the mean object query, the entropy of  $w_{h,j}$ , and the log-magnitude of  $\mathbf{v}_{h,j}^{\text{geo}}$ , and is clamped to the same range as  $s^{\text{dir}}$ .

#### 4.4. Precision-weighted fusion

Treating the two heads as independent isotropic Gaussian estimates with precisions  $\lambda_{h,j}^k = \exp(-2s_{h,j}^k)$ , the maximum-likelihood combination is

$$\hat{\mathbf{v}}_{h,j} = \frac{\lambda_{h,j}^{\text{dir}} \mathbf{v}_{h,j}^{\text{dir}} + \lambda_{h,j}^{\text{geo}} \mathbf{v}_{h,j}^{\text{geo}}}{\lambda_{h,j}^{\text{dir}} + \lambda_{h,j}^{\text{geo}}}. \quad (9)$$

This replaces the learned scalar gate of JSSR [9] and the equal-weight ensembling of GOLF [6] with a fusion rule whose weights are trained as uncertainties [47, 48] and can be inspected. We also train a learned-gate variant,  $\hat{\mathbf{v}} = (1 - g)\mathbf{v}^{\text{dir}} + g\mathbf{v}^{\text{geo}}$  with  $g = \sigma(\mathbf{w}_g^T [\mathbf{z}_{h,j}; \bar{\mathbf{z}}^o; \cdot])$ , and an equal-weight variant, as ablations.

#### 4.5. Training objective

With  $m_h \in \{0, 1\}$  marking hands with a valid target,

$$\mathcal{L} = \mathcal{L}_{\text{field}} + \lambda_{\text{dir}} \mathcal{L}_{\text{dir}} + \lambda_{\text{geo}} \mathcal{L}_{\text{geo}} + \lambda_p \mathcal{L}_{\text{joint}} + \lambda_o \mathcal{L}_{\text{obj}} + \lambda_{\text{pres}} \mathcal{L}_{\text{pres}} + \lambda_{\text{vis}} \mathcal{L}_{\text{vis}}, \quad (10)$$

where  $\mathcal{L}_{\text{field}}$  is the masked mean endpoint error of the fused field (the evaluation metric),  $\mathcal{L}_k = \frac{1}{2} \|\mathbf{v}^k - \mathbf{v}\|^2 e^{-2s^k} + 3s^k$  is the Gaussian negative log-likelihood of each head,  $\mathcal{L}_{\text{joint}}$  is an  $\ell_1$  loss on camera-frame joints,  $\mathcal{L}_{\text{obj}}$  is the symmetry-agnostic ADD-S distance [37, 38] between predicted and ground-truth posed vertices plus an  $\ell_1$  translation term, and  $\mathcal{L}_{\text{pres}}, \mathcal{L}_{\text{vis}}$  are binary cross-entropies on presence and visibility. The weights are  $(\lambda_{\text{dir}}, \lambda_{\text{geo}}, \lambda_p, \lambda_o, \lambda_{\text{pres}}, \lambda_{\text{vis}}) = (0.1, 0.1, 0.02, 0.02, 0.1, 0.1)$ , chosen once so that the auxiliary terms are of the same order as  $\mathcal{L}_{\text{field}}$  in millimetres and not tuned thereafter. All auxiliary targets are derived from the released training annotations (hand landmarks, object poses, calibration); the visibility label tests whether the ray from the reference camera to a joint intersects the posed object mesh before the joint. The joint and object terms use frames in which only the hand or only the object is confident, which the field loss cannot use: about 39% more hand instances carry a confident hand pose than a field, and 14.5% more frames carry a confident object pose than a field.

#### 4.6. Inference

Both views are encoded (or read from the cache), the decoder is run once, Eq. (8) is evaluated with the exact nearest vertex, Eq. (9) fuses the heads, and the result is rotated into the world frame with  $\mathbf{R}_{w \leftarrow c}$ . Both hands are always predicted, since abstention is penalised by the official score.

## 5. Experimental setup

### 5.1. Data and splits

We use the interaction-field release of SHOW3D [3, 5]: 468 recordings of 10 subjects manipulating 21 objects, captured at 60 fps by the two monochrome cameras of a Meta Quest 3 headset and released undistorted at  $1024 \times 1280$ . Labels for the three official test subjects are withheld and the evaluation server closed on 31 August 2026, so all results are reported on held-out *training* subjects with subject-level splits: fold A holds out XYZ109 and LYA722, the development split of the organisers' baseline and of GOLF [6], and fold B holds out MHA016 and MMO925. No recording appears in both partitions of a fold, the official test subjects are never downloaded, and an automated check verifies both properties before every run. Frames are sampled at 10 fps for label statistics, matching the organisers' protocol, and every second sampled frame (5 fps) is used for the token-based models; a frame contributes a target for a hand when headset tracking, the object pose and that hand pose are valid, exactly as in the official label code. Our derived labels reproduce the organisers' count of 87,366 labelled frames at 10 fps and match the official field to within  $10^{-14}$  mm on the frames we compared. Fold A trains on 37,645 frames and validates on 5,955 (11,948 at 10 fps, the exact development split); fold B trains on 35,052 and validates on 8,548.

### 5.2. Label statistics

Figure 4 summarises the derived labels. Of 138,579 sampled frames, 91.2% have valid headset tracking, 72.2% a confident object pose and 63.0% at least one confident hand together with the object, giving 68,270 left-hand and 79,272 right-hand fields; 60,176 frames carry both hands. The joint field is heavy-tailed (median 30.6 mm, mean 82.8 mm, 30.2% of joints beyond 60 mm) and strongly joint dependent: fingertips have a median magnitude of 13.8 mm and 52% lie within 15 mm of the object, whereas the wrist has a median of 96.5 mm and 87% of wrist vectors exceed 60 mm. Any mean-ADE number is therefore dominated by far-field vectors of the wrist, the proximal joints and non-interacting hands. Joints that the object occludes or that fall outside the image account for 24.0% of all joints and are mostly in contact (median magnitude 11.8 mm); the object is out of the reference view in only 1.3% of labelled frames and hands in 3.0%, so unobservable targets cannot account for a large share of any model's error, contrary to a hypothesis we held at the outset. The stereo baseline is 63.9 mm, the focal length 454 px and the median joint depth 308 mm (90th percentile 441 mm), giving a median disparity of 94 px and a depth resolution of 3.3 mm per pixel of disparity. Ground-truth hand landmarks carry a median error of 6.4 to 7.9 mm and object translations of 1.5 to 3.5 mm according to the dataset authors [3].

### 5.3. Metrics

We report the official mean ADE,  $\text{acc}@ \{10, 50, 100\}$  mm and recall (always 1.0 for our models, which predict both hands), and geometric measures computed with the ground-truth joints as anchors so that all methods are comparable: the surface distance

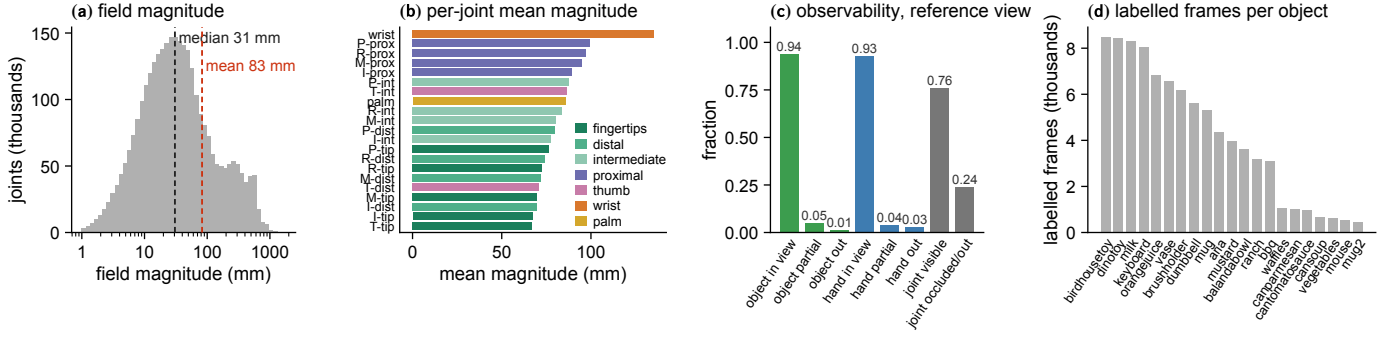


Figure 4: Statistics of the derived interaction-field labels (10 training subjects at 10 fps, 87,366 frames, 3.1 M joint vectors). (a) The magnitude distribution is heavy-tailed: median 31 mm, mean 83 mm, 30% of joints beyond 60 mm. (b) Mean magnitude per joint: fingertips and distal joints are typically in contact, the wrist is typically far from the object. (c) Observability in the reference view: the object is at least 90% inside the image in 94% of labelled frames and essentially out of view in 1.3%; 24% of joints are occluded by the object or outside the image, and those joints are mostly in contact (median magnitude 11.8 mm against 39.8 mm for visible joints). (d) Labelled frames per object; several objects are handled by one to three subjects only.

(SD) from the predicted endpoint  $\mathbf{p}_{h,j} + \hat{\mathbf{v}}_{h,j}$  to the nearest vertex of the ground-truth posed object; the direction error between  $\hat{\mathbf{v}}$  and  $\mathbf{v}$  for  $\|\mathbf{v}\| > 20$  mm; the magnitude error  $|||\hat{\mathbf{v}}|| - \|\mathbf{v}|||$ ; and the decomposition of the endpoint error into its component along the ray from the reference camera through the ground-truth endpoint and the lateral remainder. ADE is further stratified by field magnitude ( $< 15$ ,  $15$  to  $60$ ,  $> 60$  mm), by joint visibility in the reference view (inside the image and not occluded by the object, tested by ray-mesh intersection against the posed mesh), by the fraction of object vertices and of hand joints inside the reference image (at least 90%, 10 to 90%, below 10%), by joint group, hand, object and subject. For factored models we additionally evaluate each head alone and the geometric head with either argument replaced by its ground truth.

#### 5.4. Baselines

The baselines are (B1) the official SHOW3D baseline [5], which carries the InterField name of Section 2 over to an image-only ResNet-50 [16] on the full  $224^2$  frame regressing the camera-frame field; we report the organisers’ published number on the identical fold-A split rather than retraining it; (B2) a direct joint-query regressor on frozen DINOv3 ViT-B/16 tokens of the reference view; (B3) the same with both views and Plücker-ray token embeddings, i.e. the consensus architecture of the challenge solutions [6, 9] without backbone adaptation or region sampling; (B3b) B3 with the ray encoding removed; and the zero, training-mean and hand-crop-oracle reference predictors from the starter kit. FIF adds the geometric head, the auxiliary losses and the precision fusion of Section 4 on top of B3’s tokens; all token models share the encoder, the cache, the decoder width (384), the number of heads (8), the depth ( $L=6$ ), the feed-forward width (1536), dropout (0.1), token dropout (0.1), the optimiser (AdamW [60], learning rate  $2 \cdot 10^{-4}$ , weight decay 0.05,  $\beta = (0.9, 0.95)$ , 300 warm-up steps followed by a cosine schedule), the batch size (32) and early stopping (patience 3 on validation ADE with a minimum improvement of 0.05 mm), with an epoch budget of 10 that we raise to 30 for the much smaller HOT3D-only runs of Section 6.5. Nothing was tuned on the held-out subjects beyond this early-stopping rule,

which applies identically to every model. The main comparison is repeated with a second seed on fold A.

#### 5.5. Implementation

Tokens are extracted once at  $480 \times 384$  in half precision, projected by principal component analysis to 256 dimensions (the 256 components retain 92.4% of the variance of 200,000 sampled tokens) and stored as 8-bit integers with per-token scales; the cache of 124,954 images occupies 22 GB. Object meshes are the binary glTF models bundled with the dataset API; 16 control points and 1,024 vertices per object are farthest-point sampled for the geometric head, while evaluation and label generation use the full meshes. Everything is implemented in PyTorch [61] and runs on a single Apple M5 laptop with 16 GB of unified memory: feature extraction at 16 images per second, training at 48 samples per second for the stereo factored model and 102 for the single-view direct model, with a peak memory of 5.6 GB. A full training run takes 1.1 to 2.5 h and the complete set of experiments in this paper about 35 h.

## 6. Results

### 6.1. Main comparison

Table 1 gives the main comparison. On fold A, the direct regressor over frozen tokens already improves substantially on the official ResNet-50 baseline, from 60.47 to 49.58 mm with one view and to 46.15 mm with both views and ray encoding;  $\text{acc}@10$  rises from 0.150 to 0.227. This is the expected effect of a strong self-supervised encoder and confirms that the consensus challenge architecture is well reproduced at the reduced scale. The second view is worth 3.4 mm over the single view (49.58 to 46.15) and the ray encoding 1.3 mm over stereo without it (47.47 to 46.15), both larger than the 0.19 and 0.6 mm that GOLF reports at its much stronger representation [6]; the geometric cues matter more when the backbone contributes less.

The full factored model reaches 44.43 mm on fold A against 46.15 mm for the matched direct regressor, a reduction of 1.72 mm (3.7%), and 34.08 against 35.68 mm on fold B, a reduction of 1.60 mm (4.5%). The two folds agree to within 0.12 mm

Table 1: Held-out-subject results on SHOW3D. Fold A holds out XYZ109 and LYA722 (the organisers’ development split), fold B holds out MHA016 and MMO925. ADE in mm (lower is better); accuracies are fractions of joints within the threshold; SD is the surface distance of predicted endpoints to the ground-truth posed object (mm). Where two numbers are given they are seeds 0 and 1. All token models share the frozen DINOv3 ViT-B/16 cache, decoder, optimiser and schedule; the InterField row is the organisers’ published result on the identical fold-A split. Recall is 1.0 for every model.

| Method                                | Views        | Fold A (XYZ109, LYA722) |              |              |              |              | Fold B (MHA016, MMO925) |              |              |
|---------------------------------------|--------------|-------------------------|--------------|--------------|--------------|--------------|-------------------------|--------------|--------------|
|                                       |              | ADE                     | acc@10       | acc@50       | acc@100      | SD           | ADE                     | acc@50       | SD           |
| Zero-vector predictor                 | –            | 82.6                    | –            | –            | –            | –            | –                       | –            | –            |
| Training-mean predictor               | –            | 94.2                    | –            | –            | –            | –            | –                       | –            | –            |
| Hand-crop oracle box, InterField [5]  | h0           | 54.7                    | –            | –            | –            | –            | –                       | –            | –            |
| InterField ResNet-50, official [5]    | h0           | 60.47                   | 0.150        | 0.682        | 0.861        | –            | –                       | –            | –            |
| Direct regression (B2)                | h0           | 49.58                   | 0.213        | 0.742        | 0.898        | 35.75        | –                       | –            | –            |
| Direct, stereo, no ray encoding (B3b) | h0+h1        | 47.47                   | 0.205        | 0.754        | 0.908        | 34.78        | –                       | –            | –            |
| Direct, stereo + Plücker rays (B3)    | h0+h1        | 46.15 / 45.73           | 0.227        | 0.756        | 0.912        | 33.22        | 35.68                   | 0.794        | 23.61        |
| FIF, geometric head only              | h0+h1        | 55.11                   | 0.097        | 0.689        | 0.896        | 44.11        | –                       | –            | –            |
| FIF, single view                      | h0           | 45.19                   | 0.219        | 0.783        | 0.918        | 33.32        | –                       | –            | –            |
| <b>FIF (full)</b>                     | <b>h0+h1</b> | <b>44.43 / 43.82</b>    | <b>0.211</b> | <b>0.768</b> | <b>0.921</b> | <b>31.19</b> | <b>34.08</b>            | <b>0.807</b> | <b>22.75</b> |

on the size of the effect although their absolute errors differ by more than 10 mm, which reflects the difficulty of the held-out subjects rather than the models: on fold A, subject XYZ109 is at 48.7 mm and LYA722 at 38.9 mm for the factored model; on fold B MHA016 is at 42.1 mm and MMO925 at 25.4 mm. Absolute errors should therefore never be compared across folds; within-fold differences are what carries information. Across two seeds on fold A the direct model gives 46.15 and 45.73 mm and the factored model 44.43 and 43.82 mm: the gap of 1.82 mm between the seed means is three times the largest within-model seed spread (0.60 mm) and the two ranges do not overlap. With two seeds this is a range, not a variance estimate, and we report it as such.

The secondary metrics move consistently.  $\text{acc@50}$  and  $\text{acc@100}$  improve on both folds (0.756 to 0.768 and 0.912 to 0.921 on fold A; 0.794 to 0.807 and 0.920 to 0.930 on fold B), the surface distance of predicted endpoints to the true object falls from 33.2 to 31.2 mm and from 23.6 to 22.8 mm, and the magnitude error falls from 31.1 to 29.2 mm.  $\text{acc@10}$  does not improve (0.227 to 0.211 on fold A, 0.255 to 0.252 on fold B): the factored model reduces large errors and does not sharpen predictions that are already within a centimetre. The same trade appears in the single-view setting, where FIF improves on the direct model by 4.4 mm (49.58 to 45.19), two and a half times the stereo gain, and a single-view FIF at 45.19 mm is better than the two-view direct regressor at 46.15 mm. The geometric supervision helps most where the input is least informative.

Figure 5 shows held-out frames sampled across the error distribution. Where the hand grasps the object, both models are accurate and nearly indistinguishable. Where fingers hover near but not on the surface, the direct regressor tends to shorten the vectors towards the hand, whereas the factored model places their endpoints on the object. Where the hand is outside the image, both models fail, but the factored model fails by less. The failure analysis of Section 6.6 examines these regimes in turn.

## 6.2. Where the gain comes from

Table 2 and Fig. 6 locate the improvement. In every SHOW3D-only comparison the far-field stratum carries the largest gain, and in two of the three the gain grows monotonically with the magnitude of the target: on fold A stereo, 0.6 mm for joints within 15 mm of the object, 1.5 mm between 15 and 60 mm and 3.0 mm beyond 60 mm; in the single-view setting, 1.9, 2.7 and 8.7 mm. On fold B the middle stratum is flat (0.2, 0.0 and 4.5 mm). The gain is larger for joints occluded by the object or outside the image than for visible joints (3.0 against 1.3 mm on fold A) and largest by far when the hand itself leaves the image, where it reaches 21.1 mm on fold A, 22.5 mm on fold B and 19.7 mm with one view. The far-field stratum contains 32% of the joints but 67% of the total error of the direct model; it is where the mean is decided, and it is where the factored model helps most. This is the behaviour the formulation predicts: the geometric route reconstructs the field from two quantities that remain constrained when the direct mapping from pixels to a long vector is not, and the fusion falls back on the direct head when the object is in plain view and the vector is short. The second clause of the falsification criterion is therefore met, and met on two independent subject pairs.

The error components in the last block of Table 2 answer the question left open by the challenge reports, namely why a second view helps so little. For every model the lateral component of the endpoint error is larger than the component along the viewing ray, 36.7 against 22.3 mm for the stereo direct model and 39.8 against 23.4 mm for its single-view counterpart. Adding the second view reduces the along-ray component by 1.1 mm and the lateral component by 3.1 mm. The residual error of these models is therefore not primarily a depth error, which stereo could resolve, but a lateral one, which reflects where along the object surface the network believes the nearest point to be. The factored model reduces both components, by 0.5 mm along the ray and 1.6 mm laterally, and lowers the direction error of long vectors from 30.5 to 29.8 degrees. Per joint group the gain is spread across fingertips, distal and intermediate joints (about 2 mm each) and is smallest at the wrist (0.6 mm), whose vectors

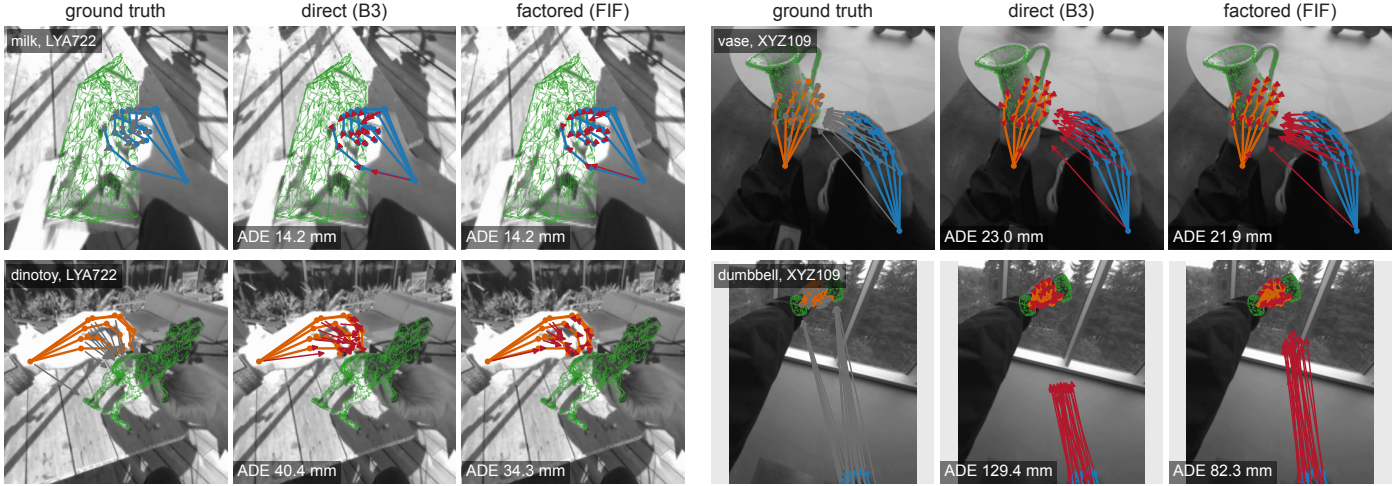


Figure 5: Held-out predictions of the direct and the factored model. The four frames lie near the 10th, 37th, 63rd and 90th percentile of the factored model’s per-frame error on fold A (top left to bottom right), one object each. Every frame is shown as ground truth (grey vectors), the direct stereo regressor and the factored model (red vectors); skeletons and the posed mesh are ground truth in every panel, light grey marks regions outside the camera image, and the per-frame ADE is printed. In the near-contact grasp of the milk carton both models are within 15 mm; on the vase and the toy the factored model corrects the direction and length of vectors from fingers that are not touching the object; in the dumbbell frame the right hand is below the image, where the direct model errs by 129 mm and the factored model by 82 mm.

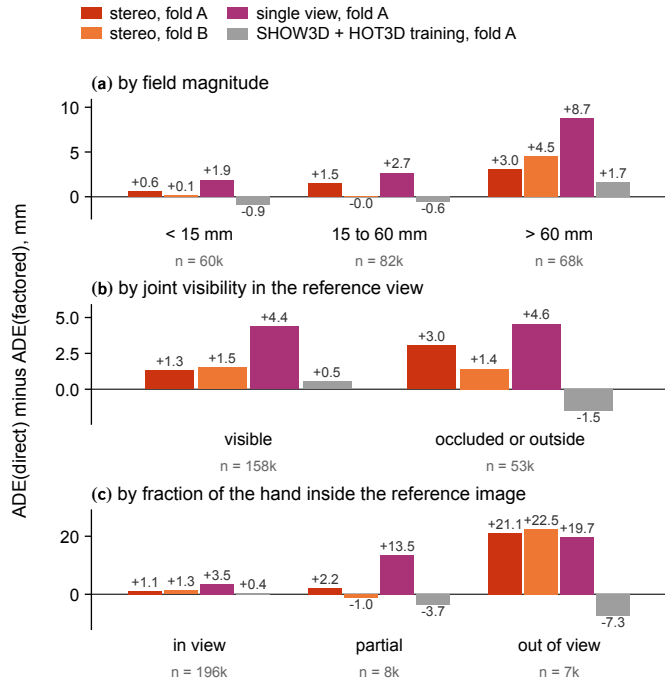


Figure 6: Where the factored model gains. Bars give ADE(direct) minus ADE(FIF) in mm per difficulty stratum for four matched pairs: stereo on fold A, stereo on fold B, single view on fold A, and joint SHOW3D+HOT3D training on fold A. (a) By field magnitude the gain grows from the near-contact to the far-field stratum in every SHOW3D-only comparison. (b) The gain is larger for occluded than for visible joints. (c) It is largest by far when the hand is outside the image. The joint-training pair breaks the pattern (Section 6.5). Sample counts are for fold A.

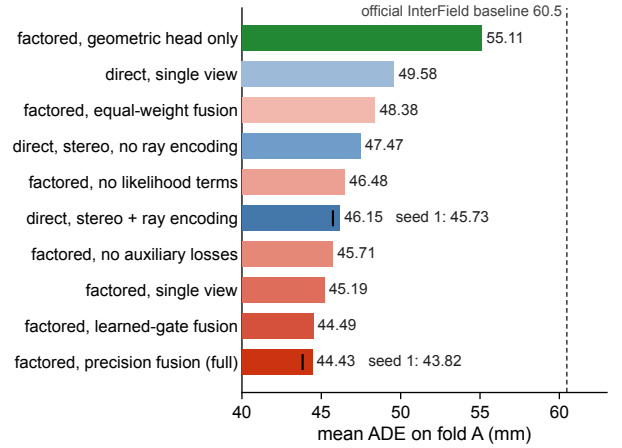


Figure 7: Fold-A held-out ADE of every trained variant, sorted. Vertical ticks mark the second seed of the two main rows; the dashed line is the organisers’ InterField result on the same split.

### 6.3. Ablations

Table 3 and Fig. 7 remove one component at a time. Three findings stand out and each corrects a claim one might otherwise be tempted to make.

*The geometric head is not competitive alone.* The geometric head by itself reaches 55.11 mm, worse than the single-view direct regressor, with acc@10 collapsing to 0.097: forced to reconstruct every field from predicted joints and a predicted pose, it inherits the errors of both arguments and adds none of the direct head’s pixel-level precision. The hybrid at 44.43 mm is better than either of its components, which is genuine complementarity, but the geometric route should not be pitched as a standalone estimator.

are the longest and least constrained by contact.

Table 2: Stratified error on fold A for the single-view and stereo direct regressors and their factored counterparts. Strata are defined from the ground truth: field magnitude, joint visibility in the reference view (inside the image and not occluded by the posed object), the fraction of hand joints and object vertices inside the reference image, and joint group. The last block decomposes the endpoint error into its component along the viewing ray through the true endpoint and the lateral remainder, and reports direction, magnitude and surface-distance errors.

| Stratum                                     |                       | $n$ (joints) | Direct, h0 | FIF, h0 | Direct, h0+h1 | FIF, h0+h1 |
|---------------------------------------------|-----------------------|--------------|------------|---------|---------------|------------|
| Field magnitude                             | < 15 mm               | 60,358       | 15.19      | 13.29   | 14.57         | 13.99      |
|                                             | 15 to 60 mm           | 82,383       | 29.09      | 26.42   | 28.21         | 26.71      |
|                                             | > 60 mm               | 68,057       | 104.81     | 96.07   | 95.87         | 92.85      |
| Joint visibility                            | visible               | 158,055      | 47.96      | 43.60   | 45.19         | 43.89      |
|                                             | occluded or outside   | 52,743       | 54.35      | 49.78   | 49.03         | 46.00      |
| Hand in view                                | $\geq 90\%$           | 195,930      | 40.61      | 37.10   | 38.34         | 37.28      |
|                                             | 10 to 90%             | 8,211        | 86.60      | 73.14   | 74.32         | 72.13      |
|                                             | < 10%                 | 6,657        | 267.16     | 247.44  | 241.36        | 220.27     |
| Object in view                              | $\geq 90\%$           | 202,944      | 45.70      | 41.51   | 41.97         | 40.51      |
|                                             | 10 to 90%             | 6,636        | 93.83      | 83.37   | 97.96         | 88.47      |
|                                             | < 10%                 | 1,218        | 451.67     | 443.08  | 460.33        | 456.28     |
| Joint group                                 | fingertips            | 50,190       | 47.28      | 43.01   | 44.16         | 42.24      |
|                                             | distal                | 40,152       | 47.22      | 42.92   | 43.89         | 41.96      |
|                                             | intermediate          | 40,152       | 49.73      | 45.27   | 46.13         | 44.22      |
|                                             | proximal              | 40,152       | 52.51      | 48.03   | 48.57         | 47.15      |
|                                             | thumb                 | 20,076       | 47.32      | 43.17   | 44.35         | 42.61      |
|                                             | palm                  | 10,038       | 48.87      | 44.06   | 45.03         | 43.47      |
|                                             | wrist                 | 10,038       | 62.97      | 57.74   | 60.22         | 59.61      |
| All joints                                  | mean ADE              | 210,798      | 49.58      | 45.19   | 46.15         | 44.43      |
| Error component                             | along the viewing ray |              | 23.43      | 21.46   | 22.29         | 21.76      |
|                                             | lateral               |              | 39.77      | 35.86   | 36.65         | 35.08      |
| Direction error ( $\ \mathbf{v}\  > 20$ mm) | degrees               |              | 31.6       | 29.9    | 30.5          | 29.8       |
| Magnitude error                             | mm                    |              | 33.22      | 29.54   | 31.07         | 29.17      |
| Surface distance                            | mm                    |              | 35.75      | 33.32   | 33.22         | 31.19      |

Table 3: Ablations on fold A. All variants share tokens, decoder, optimiser and schedule. Along-ray and lateral are the two components of the endpoint error (mm). Training time is on one Apple M5; inference time is per frame excluding the frozen encoder and including metric computation. The variant without likelihood terms cannot use precision fusion and averages the heads; the geometric-only variant has no direct head.

| Variant                                | ADE   | acc@10 | acc@50 | acc@100 | SD    | along ray | lateral | Params (M) | Train (h) | Infer. (ms) |
|----------------------------------------|-------|--------|--------|---------|-------|-----------|---------|------------|-----------|-------------|
| Direct regression, single view         | 49.58 | 0.213  | 0.742  | 0.898   | 35.75 | 23.43     | 39.77   | 14.9       | 1.1       | 9.1         |
| Direct, stereo, no ray encoding        | 47.47 | 0.205  | 0.754  | 0.908   | 34.78 | 21.84     | 38.47   | 14.9       | 1.7       | 12.0        |
| Direct, stereo + rays (B3)             | 46.15 | 0.227  | 0.756  | 0.912   | 33.22 | 22.29     | 36.65   | 15.1       | 2.2       | 12.4        |
| same, seed 1                           | 45.73 | 0.222  | 0.776  | 0.918   | 33.40 | 21.55     | 36.66   | 15.1       | 1.8       | 12.3        |
| FIF, geometric head only               | 55.11 | 0.097  | 0.689  | 0.896   | 44.11 | 25.71     | 44.31   | 15.1       | 1.9       | 21.8        |
| FIF, mean fusion                       | 48.38 | 0.173  | 0.747  | 0.911   | 32.48 | 23.28     | 38.53   | 15.1       | 2.4       | 21.4        |
| FIF, no likelihood terms (mean fusion) | 46.48 | 0.169  | 0.757  | 0.921   | 32.41 | 22.44     | 36.79   | 15.1       | 2.3       | 21.3        |
| FIF, no auxiliary losses               | 45.71 | 0.207  | 0.774  | 0.920   | 34.09 | 22.29     | 36.27   | 15.1       | 2.2       | 21.3        |
| FIF, single view                       | 45.19 | 0.219  | 0.783  | 0.918   | 33.32 | 21.46     | 35.86   | 15.1       | 1.3       | 18.2        |
| FIF, learned gate fusion               | 44.49 | 0.211  | 0.782  | 0.924   | 32.32 | 21.89     | 34.93   | 15.4       | 2.3       | 21.2        |
| FIF, precision fusion (full)           | 44.43 | 0.211  | 0.768  | 0.921   | 31.19 | 21.76     | 35.08   | 15.1       | 2.4       | 21.9        |
| same, seed 1                           | 43.82 | 0.221  | 0.787  | 0.927   | 31.87 | 21.46     | 34.50   | 15.1       | 2.4       | 22.4        |

*Adaptive weighting matters; the precision formulation specifically does not.* Averaging the two heads with equal weights gives 48.38 mm, worse than the direct regressor alone: mixing a strong estimator with a weak one at a fixed ratio destroys value. Any adaptive weighting recovers it, and the two adaptive rules we trained are indistinguishable: precision fusion reaches 44.43 mm and a learned sigmoid gate 44.49 mm, a difference of 0.06 mm, with the gate slightly better on acc@50 and acc@100. The ablation therefore supports the claim that per-joint adap-

tive weighting is necessary, and does not support the claim that likelihood-derived weights are better than learned ones. What the precision formulation offers over the gate is a calibrated per-joint uncertainty as a by-product, examined in Section 6.4.

*The likelihood terms pay only through the fusion they enable.* Removing the two Gaussian negative log-likelihood terms from the objective forces equal-weight fusion and gives 46.48 mm. Comparing this variant with the equal-weight variant that keeps the likelihood terms (48.38 mm) isolates the effect of the terms

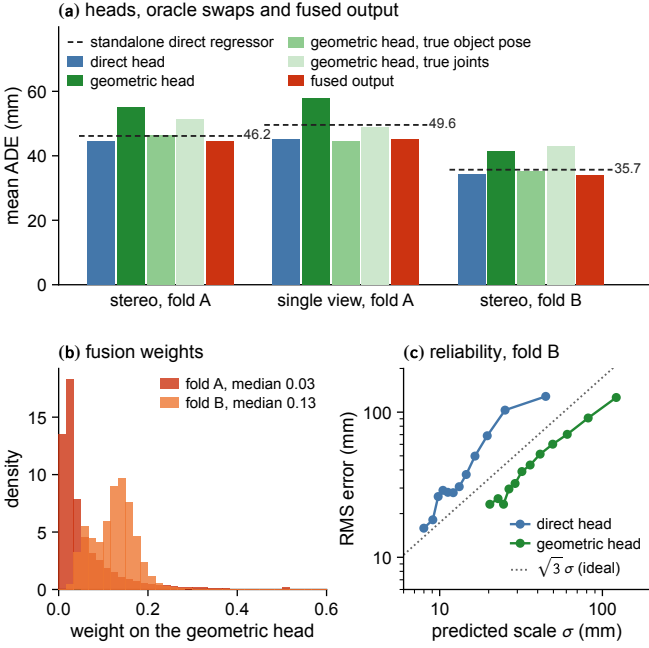


Figure 8: What each part of the factored model contributes. (a) Error decomposition of Table 4 for three factored models; the dashed line is the standalone direct regressor trained without a geometric head. The fused output tracks the direct head to within 0.2 mm, and the direct head is below the standalone baseline by 1.4 to 4.5 mm. (b) Distribution of the fusion weight on the geometric head: on fold A the geometric log-scale saturated at its clamp and the weights collapse near zero; on fold B both scales trained normally. (c) Reliability of both heads on fold B, where the geometric scale saturated at only 4% of joints: binned predicted scale against the root-mean-square error of the residual, with the  $\sqrt{3}\sigma$  line an isotropic Gaussian would follow.

at a fixed fusion rule: they make the model 1.9 mm *worse*. The likelihood objective is not a better regression loss than the plain endpoint error; its value lies entirely in producing the precisions that adaptive fusion consumes, and only the pair of likelihood terms and precision fusion reaches 44.43 mm.

*Auxiliary losses and views.* Removing the joint, object and visibility losses costs 1.3 mm (45.71 against 44.43): supervising the arguments of  $G$  helps even though only the field is evaluated. The single-view factored model at 45.19 mm loses 0.8 mm to its stereo counterpart, against 3.4 mm for the direct regressors, so the factored representation recovers most of what the second camera provides. The factored model costs 1.8 times the inference time of the direct model (21.9 against 12.4 ms per frame excluding the frozen encoder), almost entirely in the soft nearest-vertex operator over 1,024 vertices, and adds no parameters beyond the two small heads (15.1 M trainable in both cases).

#### 6.4. Inside the factored model

Because the factored model exposes its intermediate quantities, we can ask where its improvement actually comes from. Table 4 and Fig. 8 report the two heads separately, the geometric head with each argument replaced by its ground truth, and the behaviour of the learned precisions. The picture that emerges

differs from the mechanism one would assume from Eq. (9), and we state it plainly.

*The improvement is carried by the direct head.* On fold A the direct head *inside* the factored model reaches 44.55 mm on its own, 1.6 mm better than the standalone direct regressor trained without a geometric head (46.15 mm); the same holds for the second seed (43.98 against 45.73), for the single-view model (45.06 against 49.58, a difference of 4.5 mm) and for fold B (34.29 against 35.68). The fused output improves on the direct head by only 0.12, 0.16,  $-0.13$  and 0.21 mm respectively. In other words, the geometric head and the auxiliary losses act as a structured multi-task signal that shapes the shared decoder, and the direct head reaps the benefit; the combination at inference time contributes about a tenth of a millimetre. This reframes the method: what is being tested is the value of *geometric supervision* for a direct regressor, and that value is 1.4 to 4.5 mm depending on the setting.

*The object pose is the bottleneck of the geometric route.* The geometric head alone gives 55.11 mm on fold A. Replacing its predicted object pose by the ground truth lowers this to 46.46 mm, whereas replacing the predicted joints by the ground truth lowers it only to 51.35 mm: the object pose accounts for about 8.7 mm of the geometric head’s error and the joints for about 3.8 mm. Even with a perfect object pose, the geometric route (46.46 mm) does not beat the direct head (44.55 mm), because the predicted joints are themselves off by roughly the amount that SHOW3D’s own hand annotations are [3]. The object pose dominates on fold B and in the single-view model as well (41.47, 35.08, 43.11 and 58.00, 44.67, 48.79), with the difference that on fold B replacing the predicted joints by their ground truth raises the error slightly rather than lowering it. A better object-pose branch is where the geometric route has room to improve; a better joint branch has little to offer.

*The geometric scale saturated on fold A.* In the three SHOW3D-only fold-A precision-fusion runs the geometric log-scale  $s^{\text{geo}}$  sits at its upper clamp of 5 for 100% of joints, a constant predicted standard deviation of 148 mm. The optimiser found it cheaper to declare the geometric head maximally uncertain everywhere than to calibrate it, which drives the fusion weight on the geometric head to a median of 0.03 and a maximum of 0.50 and explains why the fused output is the direct head to within a fraction of a millimetre. On fold B the same architecture trained normally: the geometric scale has a median of 33.8 mm, correlates with the head’s error (Spearman 0.66) with a log-log reliability slope of 1.00 (Fig. 8c), and the median fusion weight is 0.13. The learned-gate variant, whose scales do not enter the fusion, also trained normally (median geometric scale 22.6 mm, Spearman 0.69). The saturation is thus a training pathology of the precision-fusion objective on this fold, not a property of the geometric head, and the result of Table 1 was obtained *despite* it, which is a further reason to attribute the gain to the representation rather than to the fusion. We did not intervene on the clamp once the pathology was found, because doing so on the held-out subjects would have compromised the protocol; a wider clamp or a scale-specific learning rate are the obvious remedies.

Table 4: What each part of the factored model contributes, mean ADE in mm on the held-out subjects. *Standalone direct* is the direct regressor trained without a geometric head; *direct head* and *geometric head* are the two heads of the factored model evaluated separately; *Geo., GT object* and *Geo., GT joints* evaluate the geometric head with the predicted object pose or the predicted joints replaced by their ground truth; *fused* is the model’s output. The last two columns give the median fusion weight on the geometric head and the fraction of joints at which its predicted log-scale sits at the clamp of 5 (a standard deviation of 148 mm).

| Model                     | Direct alone | Direct head | Geo. head | Geo., GT object | Geo., GT joints | Fused        | med. $w^{\text{geo}}$ | $s^{\text{geo}}$ saturated |
|---------------------------|--------------|-------------|-----------|-----------------|-----------------|--------------|-----------------------|----------------------------|
| FIF, fold A, seed 0       | 46.15        | 44.55       | 55.11     | 46.46           | 51.35           | <b>44.43</b> | 0.03                  | 100%                       |
| FIF, fold A, seed 1       | 45.73        | 43.98       | 56.72     | 45.97           | 50.14           | <b>43.82</b> | 0.03                  | 100%                       |
| FIF single view, fold A   | 49.58        | 45.06       | 58.00     | 44.67           | 48.79           | <b>45.19</b> | 0.04                  | 100%                       |
| FIF, fold B               | 35.68        | 34.29       | 41.47     | 35.08           | 43.11           | <b>34.08</b> | 0.13                  | 4%                         |
| FIF gate fusion, fold A   | 46.15        | 44.86       | 52.58     | 42.69           | 48.27           | <b>44.49</b> | 0.32                  | 0%                         |
| FIF, SHOW3D+HOT3D, fold A | 46.20        | 45.99       | 62.38     | 47.63           | 54.78           | <b>46.16</b> | 0.03                  | 100%                       |

Table 5: Calibration of the learned scales.  $\rho$  is the Spearman correlation between the predicted standard deviation and the absolute error of the corresponding head (const.: the scale is constant); *remove  $k\%$*  is the ADE of the joints that remain after discarding the  $k\%$  with the largest predicted fused scale (in parentheses: after discarding by true error, the oracle); AUSE is the area between the two sparsification curves over the first half of the joints.

| Model                      | $\rho_{\text{dir}}$ | $\rho_{\text{geo}}$ | $\rho_{\text{fused}}$ | ADE, all joints | drop 10%      | drop 50%      | AUSE |
|----------------------------|---------------------|---------------------|-----------------------|-----------------|---------------|---------------|------|
| FIF (full), fold A, seed 0 | 0.65                | const.              | 0.65                  | 44.42           | 33.41 (26.15) | 19.96 (11.77) | 7.55 |
| FIF (full), fold A, seed 1 | 0.60                | const.              | 0.61                  | 43.81           | 31.83 (24.82) | 20.66 (11.37) | 7.42 |
| FIF, single view, fold A   | 0.57                | const.              | 0.58                  | 45.15           | 33.30 (25.36) | 21.74 (11.49) | 8.56 |
| FIF, learned gate, fold A  | 0.58                | 0.69                | 0.62                  | 44.48           | 34.39 (25.35) | 21.07 (11.68) | 8.84 |
| FIF (full), fold B         | 0.63                | 0.66                | 0.64                  | 33.51           | 26.49 (22.11) | 15.29 (10.12) | 4.28 |

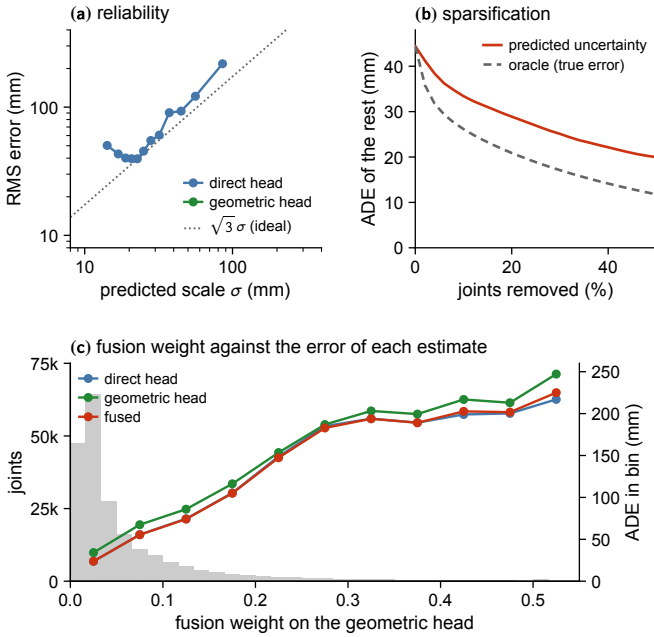


Figure 9: The learned scales as uncertainty estimates on fold A. (a) Reliability of the direct head: predicted scale against root-mean-square error in 12 quantile bins; the geometric head’s scale is constant on this fold and is shown as a single point. (b) Sparsification: discarding the 10% of joints with the largest predicted fused scale lowers the ADE of the remainder from 44.4 to 33.4 mm (oracle 26.2 mm). (c) Histogram of the fusion weight on the geometric head, and the error of the direct, geometric and fused estimates as a function of that weight: the weight rises where both heads are wrong, so the fusion does not select the better head, it signals difficulty.

*The direct head’s uncertainty is useful even where the fusion is not.* Table 5 and Fig. 9 assess the learned scales as uncertainty estimates. The direct head’s predicted scale correlates

with its error at Spearman 0.65 on fold A and 0.63 on fold B, with a log-log reliability slope of 0.96 and 1.29, and is under-dispersed by a factor of about 1.5: its root-mean-square error is 90 mm where an isotropic Gaussian with the mean predicted scale of 33.8 mm would give 58 mm, the usual over-confidence of maximum-likelihood variance heads [49]. Used to rank joints, it is informative: discarding the 10% most uncertain joints reduces the ADE of the remainder from 44.4 to 33.4 mm against an oracle of 26.2 mm, and discarding half brings it to 20.0 mm against 11.8 mm; the area between the predicted and oracle sparsification curves is 7.6 mm on fold A and 4.3 mm on fold B. Figure 9c shows what the fusion weight encodes: it rises where the direct head is uncertain, and there both heads are wrong by a similar amount, so the weight signals difficulty rather than picking a better estimator. A per-joint oracle that selected the better of the two heads would reach 40.75 mm on fold A, 3.7 mm below the achieved fusion; the geometric head is the better of the two on 28% of joints but receives only slightly more weight there (0.075) than elsewhere (0.054), nowhere near enough to select it. Closing this gap would require the geometric head’s uncertainty to be informative, which the saturation prevented.

*The error is in the tail.* Finally, the distribution of the fused error is heavy-tailed on every fold: the median joint error is 22.6 mm, the 90th percentile 85.9 mm and the 99th percentile 388 mm on fold A. The 7.9% of joints with an error above 100 mm carry 42.7% of the total error, and the 3.4% above 200 mm carry 28.7%. Any method that wants to move the mean substantially must address these frames, which Section 6.6 shows are dominated by hands that are far from the object or outside the image.

### 6.5. Cross-dataset transfer to HOT3D

**HOT3D-IF.** HOT3D [15] shares 21 objects with identical canonical meshes, the UmeTrack hand skeleton [4] and the Quest 3 headset with SHOW3D, which makes it the only public dataset on which the joint-anchored field can be generated with the same code path and no joint-convention mapping. Its Quest 3 clips differ in three respects that we resolve: they are raw fisheye streams at 30 fps, annotated with UmeTrack wrist poses and joint angles rather than landmarks, and each frame contains several objects. We rectify both streams through the Fisheye624 model (a Kannala-Brandt polynomial [62] with tangential and thin-prism terms) to SHOW3D’s virtual pinhole camera (1024 × 1280, focal length 454 px, principal point centred, rotated so that hands appear below and the left hand on the left), recover landmarks by UmeTrack forward kinematics from the per-clip hand shape (our implementation reproduces SHOW3D’s stored landmarks to within  $10^{-4}$  mm), map objects by name to SHOW3D aliases (posed SHOW3D meshes project into HOT3D’s own annotated boxes within one pixel, confirming that the canonical frames coincide), and define the target object of a hand as the nearest shared object within 150 mm. Fields, camera-frame joints, visibility and observability then follow the SHOW3D derivation exactly. Participants are split into HOT3D-train (P0002, P0003, P0010, P0013) and HOT3D-eval (P0017, P0018, P0021); training clips are the only ones with ground truth, so evaluation uses a disjoint set of training-split participants. We extract 63 clips (nine per participant, 150 source frames each, which the 10 fps stride reduces to 50) and a further 120 clips of the training participants, giving 9,150 frames of which 6,855 are labelled; at the 5 fps stride of the token models this yields 2,889 training frames and 538 evaluation frames (17,598 joint targets). The HOT3D field is lighter-tailed than SHOW3D’s (median 35.0 mm, mean 50.8 mm in the initial 63-clip build) and hands are further from the camera (median joint depth 378 against 308 mm), so HOT3D-eval is an easier set in absolute terms and only differences between models trained identically are informative.

**Zero-shot transfer.** Table 6 and Fig. 10a report the fold-A checkpoints applied to HOT3D-eval without any adaptation. Every model does better on HOT3D than on SHOW3D, as expected from its lighter tail, but by very different amounts: the single-view direct regressor improves from 49.6 to 39.0 mm, the stereo direct regressor from 46.2 to 40.2 mm, and the factored model from 44.4 to only 43.9 mm. The in-domain ranking inverts exactly: the best SHOW3D model is the worst on HOT3D, and the simplest model transfers best. The factored model is 3.7 mm behind the matched direct regressor.

**Training on HOT3D and joint training.** Training on HOT3D-train alone gives 42.6 mm (direct) and 43.3 mm (factored) on HOT3D-eval, both *worse* than the SHOW3D-trained models applied zero-shot: 37,645 frames of another dataset beat 2,889 frames of the same one. Applied in the reverse direction these models reach 78.1 and 79.6 mm on SHOW3D, barely below the 82.6 mm of the zero-vector predictor, so the small HOT3D set does not support a transferable model at all, and the transfer is asymmetric for reasons of scale rather than of domain. Adding

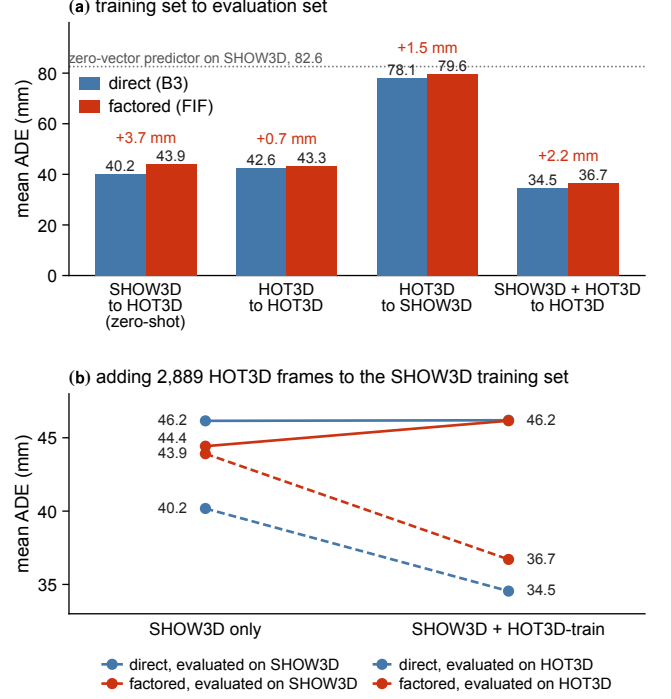


Figure 10: Cross-dataset transfer between SHOW3D and HOT3D-IF. (a) Direct against factored model in the four cross-dataset settings; the annotation is the difference (positive: factored worse). (b) The effect of adding the 2,889 HOT3D-train frames to the SHOW3D training set, evaluated on both datasets: the direct model is unchanged on SHOW3D and improves on HOT3D; the factored model loses its SHOW3D margin and improves less on HOT3D.

HOT3D-train to the SHOW3D training set is the best-powered comparison. It leaves the direct model unchanged on SHOW3D (46.15 to 46.20 mm) and improves it on HOT3D by 5.6 mm (40.2 to 34.5 mm). It costs the factored model 1.7 mm on SHOW3D (44.43 to 46.16 mm), exactly its in-domain margin, and improves it on HOT3D by 7.2 mm to 36.7 mm, still 2.2 mm behind the direct model (Fig. 10b). After joint training the two models are tied on SHOW3D and the factored one is behind on HOT3D; the stratified gains of Fig. 6 vanish as well (0.9 and 0.6 mm *against* the factored model in the near and middle strata, 1.5 mm against it on occluded joints). The third clause of the falsification criterion therefore fails in all four settings, and the advantage of geometric supervision is confined to training on a single dataset.

**Where the transfer fails.** The head-wise numbers locate the loss. On HOT3D the factored model’s direct head alone scores 42.1 mm against 40.2 mm for the standalone direct regressor, so 1.9 of the 3.7 mm are a property of the representation: the geometric supervision that helped on SHOW3D made the shared decoder more specific to SHOW3D. The remaining 1.8 mm come from the fusion, whose weight on a geometric head that now scores 63.5 mm cannot fall below the floor set by the saturated scale. Replacing the predicted object pose by the ground truth does not help the geometric head on HOT3D (63.9 against 63.5 mm), whereas it helped by 8.7 mm in domain: out of domain the predicted *joints* are the bottleneck. After joint training this reverses again (38.4 mm with the true object pose against

Table 6: Cross-dataset results (mean ADE in mm and accuracies). Each block trains on the data in the first column and is evaluated on SHOW3D fold-A validation and on HOT3D-eval. The SHOW3D block re-uses the fold-A checkpoints of Table 1; the HOT3D-train block trains on 2,889 frames and stopped early after six epochs; the joint block trains on the union. The zero-vector predictor scores 82.6 mm on SHOW3D fold A.

| Training data                 | Model                    | SHOW3D fold A |        | HOT3D-IF eval |        |        |
|-------------------------------|--------------------------|---------------|--------|---------------|--------|--------|
|                               |                          | ADE           | acc@50 | ADE           | acc@10 | acc@50 |
| SHOW3D (37,645 frames)        | Direct, single view (B2) | 49.58         | 0.742  | 38.98         | 0.202  | 0.787  |
|                               | Direct, stereo (B3)      | 46.15         | 0.756  | 40.18         | 0.201  | 0.777  |
|                               | FIF, geometric head only | 55.11         | 0.689  | 50.32         | 0.072  | 0.665  |
|                               | FIF (full)               | 44.43         | 0.768  | 43.91         | 0.183  | 0.763  |
| HOT3D-train (2,889 frames)    | Direct, stereo           | 78.13         | 0.671  | 42.63         | 0.087  | 0.750  |
|                               | FIF (full)               | 79.59         | 0.661  | 43.32         | 0.082  | 0.748  |
| SHOW3D + HOT3D-train (40,534) | Direct, stereo           | 46.20         | 0.760  | 34.54         | 0.221  | 0.818  |
|                               | FIF (full)               | 46.16         | 0.766  | 36.71         | 0.171  | 0.806  |

53.2), so the joint branch adapts to the new hands once it sees them while the object-pose branch remains the weak link. The mechanism is consistent with the ablations: the factored model’s advantage rests on the object-pose branch being useful, and that branch is the least transferable component. That models fit the dataset they are trained on is an old observation [53]; what the present result adds is that a formulation designed to inject dataset-independent geometry can *increase* that dependence when its geometric branch is trained on one distribution of object poses and meshes.

Two caveats apply. HOT3D-eval has 538 frames, so differences below about two millimetres should not be over-read, although the direction is the same in all four settings; and the HOT3D-only models early-stopped after six epochs on a noisy validation set and may be under-trained, which affects the absolute level of the HOT3D-train block but not the joint-training comparison.

### 6.6. Failure analysis

Figure 11 shows the worst held-out frame of the factored model in each of four categories, and the category statistics explain where the mean of Table 1 is decided. We group the 5,955 fold-A validation frames by properties of the ground truth, computed once and applied to both models.

*Non-interacting hands dominate the tail.* In 1,152 frames (19% of the validation set) at least one hand has a median field magnitude above 200 mm: the hand is not interacting with the object at all, and the label is the vector to an object the person is holding in the other hand, has put down, or is about to pick up. On these frames the factored model averages 102 mm and the direct model 105 mm. The 32 frames in which the object is essentially out of the reference image average 426 and 429 mm. The single worst frames are all of one kind: subject XYZ109 types or waves with one hand while the mouse rests far away, and the true field is a vector of more than one metre that both models miss by the same amount (Fig. 11, bottom right). Such frames are legitimate under the task definition but carry no information about hand-object interaction; whether they should count towards an interaction-field metric is a question for the benchmark rather than for the models.

*One unseen object accounts for a sixth of the error.* The mouse is handled by a single subject in the training release, and that subject (XYZ109) is held out in fold A, so the fold-A validation set tests the mouse as an unseen object with 11,130 joint targets at 134.5 mm (direct: 136.3 mm). Its contribution to the mean, the product of its share of joints and its stratum error, is 7.1 mm of the factored model’s 44.4 mm; the dumbbell, whose exercise sequences move the free hand far from the weight, contributes a further 10.9 mm at 65.1 mm over 35,427 joints. Two of nine objects therefore account for 18 of the 44 mm. The remaining seven objects lie between 24.5 mm (vase) and 50.2 mm (orange juice), and the factored model improves on the direct one for all but two of them (orange juice by 3.7 mm and dinotoy by 0.1 mm in the wrong direction, birdhousetoy by 5.3 mm in the right one); on the two heavy contributors it gains as well, 2.8 mm on the dumbbell and 1.8 mm on the mouse. On fold B, where the held-out subjects handle twelve objects that all appear in training, the object errors range from 19.4 to 51.4 mm and the factored model improves on ten of the twelve.

*Occlusion by the object is not where the gain is.* The 3.0 mm gain on the *occluded-or-outside* stratum of Table 2 might suggest that the geometric route helps when fingers are hidden behind the object. It does not. Take the 812 frames in which a hand is inside the image but more than half of its joints are occluded by the posed mesh and its median field magnitude is below 15 mm, that is, the hand is wrapped around the object. On those frames both models score 29 mm (factored 29.3, direct 29.2). The stratum gain comes from the joints *outside* the image, where the factored model is 21 mm better. For joints within 15 mm of the object, moreover, both models are barely better than predicting the zero vector (14.0 and 14.6 mm against a bound of 15 mm): at contact the models have learned that the field is small but not which small vector it is, and the object’s own annotation noise of 1.5 to 3.5 mm [3] is a floor on what can be recovered there.

*Hands outside the image.* When a hand is below 10% inside the reference image (311 frames), the factored model averages 152 mm and the direct model 165 mm. Because the hand is not observed, both models extrapolate from the other hand, the object and the body; the factored model’s advantage of 21 mm

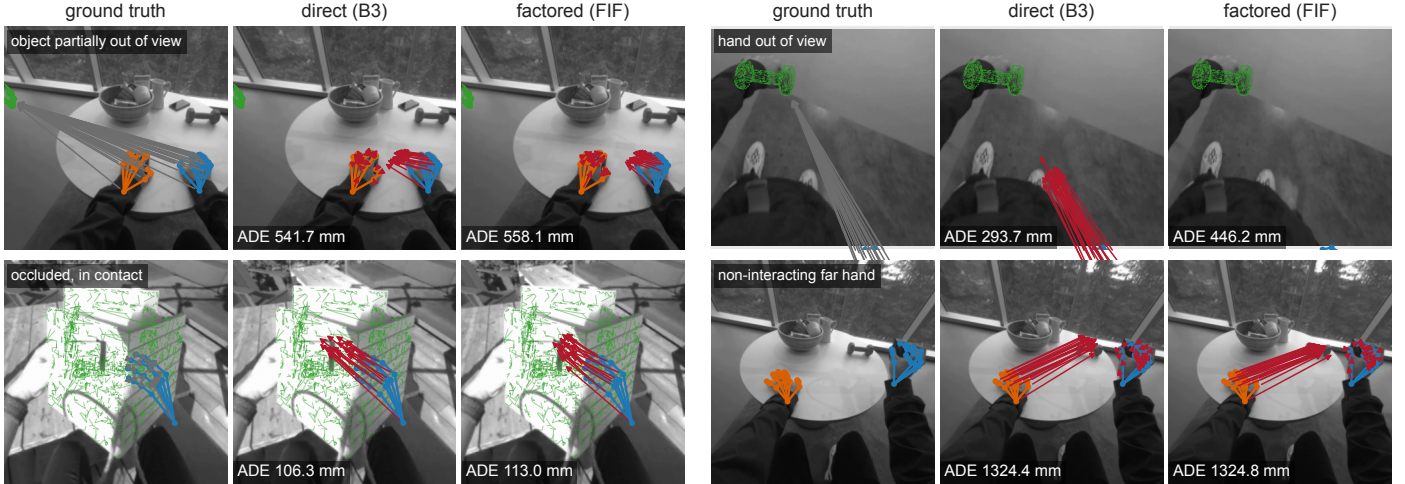


Figure 11: Failure cases: the worst fold-A frame of the factored model in four categories of difficulty, with the direct model on the same frame. Ground-truth vectors are grey, predictions red; the crop is chosen to contain the true endpoints and light grey marks regions outside the image, so a skeleton drawn there is a hand that the camera does not see. Top left: the object (green, left edge) is almost out of the image and the hands are far from it. Top right: the right hand is below the image; the direct model still anchors its vectors, the factored model does not. Bottom left: the hand grips the birdhouse and most of its joints are hidden behind the object; both models err by more than 100 mm on a hand whose true field has a median magnitude below 15 mm. Bottom right: the right hand rests on a mouse while the left rests 1.3 m away; the true field of the left hand is a 1.3 m vector that no model predicts.

on this stratum is the largest of any and is consistent across folds and views (Fig. 6c). The worst individual frame in this category (Fig. 11, top right) is one where the direct model still anchors its vectors to the invisible hand and the factored model does not, so the gain on the stratum is a property of the average, not of every frame.

*Clean frames.* On the 896 frames in which the object is in view, both hands are inside the image, at least 80% of joints are visible and the median field magnitude lies between 15 and 60 mm, the factored model scores 30.4 mm and the direct model 32.7 mm. The difference on these frames (2.3 mm) is larger than the overall difference (1.7 mm) in absolute terms and much larger in relative terms (7%), which is the cleanest statement of what geometric supervision buys when the scene is well observed.

## 7. Discussion

*What the criterion decided.* Of the three clauses fixed before training, the first two hold and the third does not. The factored model beats the matched direct regressor on both held-out subject pairs and on both seeds, by 1.6 to 1.8 mm with two views and by 4.4 mm with one; the gain is concentrated where the formulation predicts, in far-field, occluded and out-of-view joints, and the far-field stratum carries the largest gain in every single-dataset comparison. It does not transfer to a second dataset, in any of four settings, and it disappears when a small amount of that dataset is added to training. The hypothesis that a factored representation helps is therefore not rejected, but its scope is narrower than the one we set out to test: factoring helps a regressor trained on one distribution of objects and poses, and the help is largest where the direct mapping is least constrained.

*The gain is in the representation, not in the fusion.* The analysis of Section 6.4 changes how the result should be read. We designed FIF as two estimators combined by their precisions, expecting the geometric head to take over where the direct head is unsure. What happened is different. The geometric head, trained on the same tokens, is 11 mm worse than the direct head; its predicted scale saturated on fold A; the fusion weight on it is a few percent; and the fused output is the direct head to within 0.2 mm on every fold and seed. Yet the direct head inside the factored model is 1.4 to 4.5 mm better than the same head trained without the geometric branch. The mechanism is auxiliary supervision through structure: forcing the shared decoder to produce joints and control points from which a differentiable operator can rebuild the field gives the direct head a better representation to regress from. This is a more modest claim than the one the architecture suggests, and a more useful one, because it identifies the component that can be removed at inference time. A practitioner who wants the gain without the 1.8-fold inference cost of the soft nearest-vertex operator can train with the geometric branch and deploy the direct head alone.

*Why the fusion did not add more.* The per-joint oracle that picks the better head reaches 40.8 mm on fold A, 3.7 mm below what the fusion achieved; the geometric head is the better one on 28% of joints. The complementarity exists and was not exploited, for two connected reasons. The geometric head’s scale collapsed to a constant on fold A, so the fusion weight carried information about the direct head only; and where the direct head is uncertain, the geometric head is usually wrong as well (Fig. 9c), so the weight signals difficulty rather than a preference. On fold B, where both scales trained normally and the geometric weight has a median of 0.13, the fusion still added only 0.2 mm. Realising the oracle gap would require a geometric head that is right on a different set of joints from the direct head, not merely a better-

calibrated one, which points at the object-pose branch: with the true object pose the geometric head is within 2 mm of the direct head.

*What limits the geometric route.* Two numbers bound it. In domain, replacing the predicted object pose by the truth lowers the geometric head’s error by 8.7 mm and replacing the predicted joints lowers it by 3.8 mm; the predicted joints are already within the noise of the dataset’s own hand annotation. Out of domain the ordering reverses, and the joint branch, not the object branch, is what fails first. A better object-pose branch is where the geometric formulation has headroom on SHOW3D; a joint branch that generalises across capture systems is what it would need to transfer. Neither is specific to interaction fields; both are the standard problems of hand-object pose estimation [32, 30], which the factored formulation inherits along with their solutions.

*Relation to the challenge systems.* Our stereo gain of 3.4 mm is eighteen times the 0.19 mm GOLF measured at a LoRA-adapted ViT-H+, and our ray-encoding gain of 1.3 mm is twice its 0.6 mm [6]. Our error components show that even at our representation the residual is mostly lateral rather than along the ray. Both observations say that the value of geometric structure shrinks as the backbone improves, because a stronger representation already encodes it. We expect the same of the factored gain: at the scale of the winning entries it may be smaller than the 1.7 mm reported here, and an unadapted ViT-B is the setting in which it is easiest to measure. The result is a controlled measurement of one design decision, not a leaderboard entry, and the absolute numbers should be read accordingly.

*Where the remaining error is.* Independently of the method, the analysis makes three statements about the benchmark that any future system will face. First, the mean is decided in the tail: 8% of joints carry 43% of the error, and the worst of them are hands that are not interacting with the object at all, for which the label is a vector of up to 1.3 m. Second, one object handled by one subject accounts for a sixth of the fold-A error, so subject-level splits on this release test object generalisation as much as subject generalisation. Third, at contact both models are barely better than predicting the zero vector, and the object annotation noise is a floor there. A benchmark that separated interacting from non-interacting hands, or reported the field only for hands within some distance of the object, would measure the quantity its name promises more directly; the stratified numbers we release allow that separation post hoc.

## 8. Limitations

Our study is deliberately controlled and correspondingly narrow. All token-based models use a frozen DINOv3 ViT-B/16 whose features are compressed to 256 dimensions and 8 bits; GOLF’s hidden-test result relies on LoRA-adapted ViT-H+ features at higher resolution and on region sampling, which we do not reproduce, so absolute errors are not comparable to the leaderboard and our claims concern differences at a fixed representation. The evaluation server is closed and the test labels

are withheld, so all numbers are on held-out training subjects; the official baseline is compared through the organisers’ published number on the identical split rather than a retrained model. Two folds and two seeds for the main comparison give a range, not a variance, and the ablations and the second fold are single runs; differences below about half a millimetre should not be interpreted. The geometric head’s predicted scale saturated on fold A and we did not intervene, because adjusting the clamp after inspecting held-out results would have compromised the protocol; a wider clamp, a warm start of the scale head or a separate learning rate are untested remedies. The geometric head presupposes a known rigid mesh and a given category; articulated or unknown objects would require a learned surface, and a category classifier would only partially close the gap. Our visibility labels ignore occlusion by the other hand. Ground-truth hand landmarks and object poses are themselves estimates with median errors of 6 to 8 and 1.5 to 3.5 mm [3], which bounds what can be measured at contact. HOT3D-eval has 538 frames from three participants. Temporal context is available at test time and unused. Every experiment ran on one 16 GB laptop, which set the scale of the study; the same code runs unchanged on larger hardware.

## 9. Conclusion

The joint-anchored interaction field of SHOW3D is a deterministic function of the hand joints, the object pose and a known mesh, and this paper asked what is gained by predicting those arguments and applying the function rather than regressing the field directly. Under a protocol fixed in advance, at a frozen representation shared by every model, the factored formulation lowers the held-out error of a matched direct regressor from 46.2 to 44.4 mm on the official development subjects and from 35.7 to 34.1 mm on a second pair, with two seeds whose ranges do not overlap, and a single-view factored model outperforms the two-view direct one. The gain grows with the difficulty of the target, from under a millimetre at contact to 21 mm when the hand leaves the image. Opening the model showed that the gain is carried by the direct head and delivered through the shared representation: the geometric branch is a structured auxiliary task, inference-time fusion contributes a tenth of a millimetre, and the branch can be dropped at deployment. The object pose is the bottleneck of the geometric route in domain and the joints are the bottleneck out of domain, and the advantage does not survive a change of dataset or the addition of a second one to training. We release the code, the derived labels, the HOT3D-IF protocol, every checkpoint and the per-frame predictions so that these measurements can be repeated at stronger representations, where we expect the effect to be smaller and the analysis to be the same.

## CRedit authorship contribution statement

**Tanvir Ahmed:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing (original draft, review and editing), Visualization.

## Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Declaration of generative AI in the writing process

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## Data availability

SHOW3D and HOT3D are public datasets released by their authors under their respective licences (CC BY-NC 4.0 and CC BY-SA / CC BY-NC-SA); neither is redistributed here. The code, the configurations of every run, the HOT3D-IF build and evaluation protocol, the per-run metrics from which every table and figure in this paper is generated, and the scripts that generate them are available at <https://github.com/minhaz-42/factored-interaction-fields>. The derived interaction-field labels, the trained checkpoints and the per-frame predictions in the official line-delimited JSON schema are too large for that repository and are archived separately, with the location given there.

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